

Research Methods in Astrophysics

Due Feb 26, 2026

Black Hole Mass Correlations

In this project you will use halo catalogs from the IllustrisTNG simulations to measure the relation between (total) black hole mass and other galaxy properties. The first part of this project will require you to read the online documentation for the data products and identify what data you need from the provided tables to make your measurement. Once you have identified and collated the necessary data, you will plot and measure the relation between black hole mass and other galaxy properties.

1. Background

The main site that distributes Illustris data products and supports their use is [tng-project.org](https://www.tng-project.org). Particularly important for this project are the data access overview page (<https://www.tng-project.org/data/>) and the data specification page (<https://www.tng-project.org/data/docs/specifications/>). You should peruse these to get a sense of what data is available and what you may want to make use of. There are also visualization and exploration tools you may be interested in.

Relevant Papers

Nelson et al. 2019 - First Results from the TNG50 Simulation: Galactic outflows driven by supernovae and black hole feedback. <https://arxiv.org/pdf/1902.05554.pdf>

Pillepich et al. 2019 -- First results from the TNG50 simulation: the evolution of stellar and gaseous discs across cosmic time. <https://arxiv.org/pdf/1902.05553.pdf>

For scientific content regarding black hole growth, we recommend reading section 14.2 of of *Galaxy Formation and Evolution* by Mo, van den Bosch & White. This text is available in eBook format on the UVA library website. This [link](#) should prompt you to login with your UVA credentials and then allow you to download the reading.

2. Software Setup

To download data from [tng-project.org](https://www.tng-project.org) you will need to sign up for an account. **You should do this very soon after this project is assigned** as there may be a wait time before your account is activated and you are able to access the necessary data. If you have issues making an account or getting approved for API access, inform your instructor and they may be able to provide you an alternate method of acquiring the data.

Note: It is possible to work with the Illustris data without downloading local files via their web API and cloud JupyterLab integration. Students have used this setup successfully in the past, but it can be difficult for instructors to troubleshoot cloud-specific problems/bugs. Information on the cloud JupyterLab setup is available [here](#). I will describe below some of the steps necessary for downloading Illustris data for local analysis.

For local analysis you will need the `illustris\_python` package available at https://github.com/illustristng/illustris_python. On Linux this can be installed as

- `git clone https://github.com/illustristng/illustris_python.git`
- `cd illustris_python/`
- `pip install --user .`

if you use pip as your package manager for python. There are similar packages for other programming languages available under the `illustristng` GitHub organization. Examples of how to use this package to analyze the Illustris data products are available at <https://www.tng-project.org/data/docs/scripts/>. **This documentation is very useful; you should review it before attempting to move to the analysis in Section 3.**

Once your account has been activated and you have API access, navigate to <https://www.tng-project.org/data/downloads/TNG100-1/>. Scroll down the snapshot list to snapshot 99; this is the last saved output for the simulation at present day (redshift $z=0$). Click on the `[Groupcat] (4.1 GB)` icon in the second to last column and it should present you with a `wget` command that will download the group catalogs for this snapshot when run from your terminal. You should also download the offset file in the next column to the right. These total ~5 GB of data.

If the 5 GB data volume is problematic, you may use TNG100-2 or TNG100-3 which are lower resolution simulations with much lower data size. For example, the same files from TNG100-3 (<https://www.tng-project.org/data/downloads/TNG100-3/>) are little more than 100 MB.

For the `illustris\_python` package to work properly, your directory structure for the data should end up looking something like the following,

- TNG100-1
 - groups_099
 - fof_subhalo_tab_099hdf5
 - postprocessing
 - offsets
 - offsets_099.hdf5

The script examples webpage at <https://www.tng-project.org/data/docs/scripts/> has further information.

If you spend more than 2 hours trying to set up your working environment, contact your instructor / TA for assistance.

3. Simulation Analysis

You should review the data columns available in the halo catalogs you downloaded and identify the columns most suitable for examining the relation between black hole mass and dark matter halo mass. Descriptions of the data columns can be found on the data specification page (<https://www.tng-project.org/data/docs/specifications/>) in section 2 describing the group catalogs. You should additionally identify the difference between the friends-of-friends (FoF) and SubFind entries, which are accessible via the `loadHalos` and `loadSubhalos` functions in `illustris\_python`, respectively. It may be helpful to read section 2 of Rodriguez-Gomez et al. 2015 (<http://adsabs.harvard.edu/abs/2015MNRAS.449...49R>).

Step 1: After identifying the data columns you wish to use, plot the relationship between dark matter halo mass (x-axis) and black hole mass (y-axis) **for both subhalo (SubFind) and group (FoF) quantities**. These plots must be included in the assignment you turn in, in addition to a short description of which data columns you used and why. If you performed any additional data cleaning or alteration you should summarize it here as well. Given the data volume you should think about how to present this data most clearly.

Step 2: Come up with a reasonable physical equation relating the black hole mass to the halo mass. You do not need to estimate physical constants, but you should explain why you chose the scaling that you did. Hypothesize as to the physical mechanisms that could give rise to such a relationship between these parameters.

Step 3: Determine if there is a difference in the relation between the SubFind and FoF catalogs. Form a hypothesis as to why there should or should not be a difference and include it in your write-up.

Step 4: Plot the relation between galaxy stellar mass and black hole mass and comment on it briefly. Is it tighter or more scattered than the halo mass – black hole mass relation? Is the correlation what you expected? And so on. You may provide other supporting plots here if you wish.

Step 5: Explore the data columns to identify another column that is correlated with black hole mass. Plot the relation and hypothesize as to the origin of the correlation.

Step 6: Comment on any limitations of the data products that you believe may impact the above relations. If your work were to be published in a journal paper, what concerns would you have?

ASTR 5140 Only:

1. Make cuts on the halo catalog to identify a list of subhalos with well-defined black hole masses and velocity dispersions (SubFind field `[`SubhaloVelDisp`]`). Summarize your cuts and plot this relation. Fit a model of your own design to this relation via least-squares or other similar technique. Comment on your result.
2. Review section 14.2 of *Galaxy Formation and Evolution* by Mo, van den Bosch & White – see the “background” section above for a link to read the ebook on the UVA library website. Assume all black holes accrete at their Eddington luminosities and compute the corresponding AGN luminosities for the galaxies you selected from IllustrisTNG.
 - a. Calculate and plot the average AGN luminosity as a function halo mass.
 - b. Plot the distribution (histogram) of AGN luminosities.
 - c. What fraction of halos in TNG100-1 have Eddington AGN luminosities (as computed above) greater than $1.28e46$ ergs / s?
3. Now examine the actual black hole accretion rates as listed in the TNG catalogs.
 - a. Make a plot comparing the black hole accretion rates in the TNG catalogs to the Eddington mass accretion rate as a function of halo mass. Discuss whether the TNG results are consistent with Eddington accretion.
 - b. Let the efficiency of converting mass accretion into luminosity (ϵ in the reading) be 0.1. Compute AGN luminosities for the galaxies you selected from IllustrisTNG using the black hole accretion rates in the TNG catalogs. Plot the average AGN luminosity

calculated this way as a function of halo mass and compare to the same plot you made in 2a) above assuming Eddington mass accretion.

- c. Plot the distribution of AGN luminosities and compare to the plot from 2b) above.
- d. What fraction of halos in TNG100-1 have AGN luminosities greater than 1.28×10^{46} ergs / s when using the TNG mass accretion rates? How does this compare to your answer from 2c) where you assumed Eddington accretion? Comment on how the assumption of Eddington accretion rates may affect predictions for the frequency of observing bright quasars.